ISBN 979-8-3195-4817-7

NAVMATE: Design and Implementation of an Augmented Reality Based Indoor Navigation System Using GPS - Compass Sensor Fusion

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2026 International Conference on System, Computation, Automation and Networking (ICSCAN) | 979-8-3195-4817-7/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICSCAN66520.2026.11588378

***Abstract*— In large buildings like colleges, hospitals, and shopping malls, indoor navigation is a persistent problem. For example, existing location-based services can only provide navigation up to the main entrance of a building, then there is no longer guidance to get you to a specific room. This paper presents NAVMATE, a Flutter-based Android application that provides indoor navigation using augmented reality technology based on GPS and compass sensor fusion. NAVMATE uses the device's camera to overlay a real-time directional arrow onto the camera view using the Haversine formula to calculate a bearing direction, with 30% linear interpolation smoothing for stable AR tracking. NAVMATE comes equipped with a 3D campus map, multi-floor support, and automatic floor transition detection, and it also features an AI voice assistant for hands-free room navigation. Offline mode is provided by caching the building data to the device's local storage. In addition, NAVMATE has a role-based administrative interface that allows for management of building/room information. Through experimental evaluations of NAVMATE, a 92% navigation accuracy was achieved a 94% tracking stability, and the overall positional drift was 69% lower than existing systems. Since there is no requirement for additional hardware required to use NAVMATE, it is both practical and scalable as an indoor wayfinding solution.**

Keywords*— *Augmented Reality, Indoor Navigation, GPS-Compass Sensor Fusion, Haversine Formula, Flutter, Firebase Cloud Firestore, Multi-Floor Navigation, Voice Assistant, Offline Mode.

I. INTRODUCTION

Many first-time visitors and new users to large and complex buildings (such as a college campus, hospital, shopping mall, or office building), have difficulty finding their way to the correct department or room. Although GPS-aligned mobile navigation applications (Google Maps, etc.), have fundamentally changed the way people navigate outdoors, they do not provide room-to-room navigational data in large buildings simply because building material or structural designs typically block or severely distort GPS signals [5].

In addition to being an inconvenience to new students trying to find their way around classroom buildings, to patients attempting to locate hospital departments, and to visitors working their way through large office buildings, having no means to find one's way throughout a building is also a loss of time for all three groups of individuals. The majority of the current methods available for indoor navigation (printed maps, bulletin boards, and manual directions) are static; unreliable; and unable to be scalable to larger buildings[12]. Most existing AR-based solutions (like Google AR Core, and Google Scene form) generally require either that you own high-end devices or possess specialized hardware; therefore, they are often prohibitive for widespread use and adoption [1], [14]. Systems based upon BLE beacons can provide consumers with a relatively high degree of localization accuracy, however, they also require expensive hardware to be installed and an ongoing commitment for overall maintenance throughout every level of a large building [2], [8]. In addition, although WiFi fingerprinting requires extensive manual data collection prior to training model, the fingerprinting method will require frequent retraining of the model making it impractical for any dynamic environment. [3], [10].

QR code solutions mainly provide static printed navigation support, lacking real-time navigation support through the printed markers [4]. While GPS-compass fusion techniques have been studied, the vast majority of these are confined to single floor navigation, and many do not provide voice-assisted interaction, offline support, or dynamic building management features [6], [7], [15]. To address these limitations, the research presented here provides a solution called NAVMATE, an Android mobile application created via Flutter, to enable indoor navigation through Augmented Reality (AR) and the fusion of GPS compass sensors, multi-floor navigation, artificial intelligent (AI) voice-enabled assistant interaction, and offline support. The admin user will be able to register buildings and create individual room entries by entering their GPS coordinates. The admin will also use the Admin Panel to manage events and building data on the buildings registered in NAVMATE. Users will be able to

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search and identify individual rooms, and the system will provide real-time AR assistance via the smartphone's camera and directional arrows on different floors. The system will also allow for voice command input and will function without wireless communication through locally cached building data. The solution is usable on commercially available standard Android devices; additional hardware is not required for operation.

II. LITERATURE REVIEW

Over the past 10 years, researchers have looked at many different technologies for getting around indoors, including augmented reality, GPS-compass integration, Bluetooth Low Energy (BLE) beacons, Wi-Fi fingerprinting, and QR code systems. This work has produced some very interesting technologies, but many of them still suffer from significant problems related to cost, scalability, and ease of use; these issues led to the creation of NAVMATE.

Kargl et al. [1] introduced the iNAV indoor navigation system, which was one of the earliest examples of a system built to provide indoor location service capabilities. While the iNAV was capable of providing some basic navigation capabilities, it was dependent on having predefined maps of buildings and did not include any modern smartphone-based augmented reality technology.

Alnabhan and Tomaszewski. [2] developed an augmented reality-based indoor navigation system called INSAR, which overlays navigation information onto the camera view on a smartphone. Although this provided for an improved user experience through augmented reality, it did not support dynamic building operations or offline navigation, and was reliant on having designated infrastructure prior to moving to a location.

Lo et al. [3] investigated the capability of utilizing intelligent mobile devices for indoor wireless location determination and augmented reality based navigation. Their results showed that a smartphone based navigation system could be feasible but would require an extensive number of signal measurements and supporting infrastructure to accurately position the device. Qu et al. [4] provided an overview of multi-sensor fusion techniques used in indoor navigation. They demonstrated that by using multiple sensor technologies together, the accuracy of the localization process could be improved; however, implementing such systems could lead to an increase in computational complexity and cost of the overall system.

Nguyen et al. [5] created a smartphone based indoor tracking system to estimate the user's location based on Wi-Fi and BLE signals across several floors. Their research found that the tracking system improved the accuracy of the localization process; however, it required a dense deployment of wireless infrastructure for continuous monitoring of the signal.

Adapa et al. [6] navigation system using augmented reality with the help of visual markers and SLAM for visually impaired individuals. The major limitation with this system will be whether or not one can mark a path using visual markers that are visible to the user. While improving the accuracy of the user's location, the system has very limited expansion or scalability opportunities for indoor navigation due to needing additional visual markers and specialized camera configurations.

Parhizkar et al. [7] also discussed an AR navigation system for mobile devices using GPS and augmented reality. The advantages of this system over the aforementioned indoor navigation system as well as other systems designed for indoor navigation were that they improved the user experience; however, it found significant deterioration indoors due to the degradation of GPS signals within most buildings.

Patil et al. [8] created a smartphone application for indoor navigation using GPS and mobile sensors for users. This application eliminated a lot of infrastructure requirements; however, this application does not have features like augmented reality navigation, multi-floor navigation, or navigation assistance via voice. Hsu et al. [9] investigated the use of smartphones to enable indoor localization and navigation through various methods.

They found that using smartphone sensors could enhance the accuracy of indoor localization, however, could also lead to drift errors and would be limited in their scalability. Al-Bashiri et al. [10] implemented an AR-based indoor navigation system (ARN) using low energy Bluetooth (BLE) beacons as an accurate localization method. While ARN achieved higher levels of positioning performance than other methods of indoor navigation, it was found that installing and maintaining a BLE beacon system required substantial financial costs and human resources.

Zafari et al. [11] presented an exhaustive review on many of the types of systems and technologies used for indoor localization. It was concluded in this article, that much of indoor navigation relied on the use of infrastructure such as, BLE beacons, Wi-Fi access points or RFID systems which can add significant financial burdens, in terms of both deployment and ongoing maintenance costs. Bahl and Padmanabhan. [12] the RADAR indoor positioning system which incorporated the use of received signal strength measurements, for indoor localization. This system was one of the first of its kind and therefore required significant calibration and wireless infrastructure requirements.Mur-Artal et al. [13] the ORB-SLAM visual SLAM system is an accurate localization solution using computer vision techniques, but the computational demand of SLAM systems can create difficulties for real-time realisation of mobile devices.

Liu et al. [14] provide an overview on many wireless indoor positioning systems, and highlight issues of signal interference, dependency on infrastructure, and variations in accuracy because of the complexity of indoor environments. High Tower and Borriello [15] talk about the challenges facing location systems for ubiquitous computing and the need for scalable, low-cost and infrastructure-independent solutions for localisation.

TABLE I. LITERATURE REVIEW SUMMARY

Year	Authors	Method	Limitation
2007	Kargl et al. [1]	iNav indoor navigation for location based services	Static building data, no AR
2014	Alnabhan &	INSAR AR indoor nav using	No dynamic building management,

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Year	Authors	Method	Limitation
	Tomaszewski. [2]	smartphone camera guidance	limited scalability
2010	Lo et al. [3]	Mobile device based indoor tracking with AR visualization	Requires signal infrastructure and calibration
2021	Qu et al. [4]	Multi-sensor fusion methods for indoor navigation	High computational complexity
2021	Nguyen et al. [5]	Smartphone based indoor tracking with WiFi and BLE signals	Infrastructure dependency
2024	Adapa et al. [6]	Visual assist indoor navigation using visual markers and SLAM	Requires additional visual markers
2013	Parhizkar et al. [7]	Mobile AR nav using GPS based positioning	Gps unreliable in indoor environment
2019	Patil et al. [8]	Smartphone indoor nav using GPS and sensors	No AR overlay or multi floor navigation
2018	Hsu et al. [9]	Indoor localization using smartphone sensor	Sensor drift and limited accuracy
2020	Al-Bashiri et al. [10]	ARBIN AR indoor nav using BLE beacons	Hardware installation and maintenance
2019	Zafari et al. [11]	Survey of indoor nav technologies	Infrastructure heavy solution
2000	Bahl & Padmanabhan [12]	RADAR RF indoor positioning system	Requires calibration and infrastructure
2015	Mur-Artal et al. [13]	ORB-SLAM visual localization and mapping	High computational requirements
2007	Liu et al. [14]	Survey of wireless indoor positioning systems	Signal interference and accuracy issues
2001	Hightower & Borriello [15]	Location systems for ubiquitous computing	Scalability challenges

III. PROPOSED SYSTEM

NAVMATE is a Flutter-based Android application that allows for indoor navigation using augmented reality and a GPS-compass sensor fusion method. It offers multi-level navigation options, voice command capabilities, and stand-alone functionality. NAVMATE can be installed on most standard Android devices with no additional hardware or infrastructure required, making it an efficient and expandable solution for larger facilities, including college campuses, shopping malls, and hospitals [2],[7].

A. AR Real-Time Navigation

The aim is to build a real-time AR navigation application for users, displayed as an arrow superimposed on the video feed from the device's camera. The AR navigation arrow'sdirection will be determined by thefusion of the GPS location and the compass direction sensors to guide users from their current location to any registered building room without using ARCore or any special hardware [2], [7].

B. GPS/Compass Fusion Algorithm

The bearing will be calculated using the Haversine formula which calculates the distance between two points on the earth from their longitude and latitude. The compass direction will be used to rotate the AR arrow to point the user in the correct direction. The movement of the AR arrow is then smoothed/modified using 30% linear interpolation [4], [11].

C. Multifloor Navigation

NAVMATE will give you the ability to navigate within a building on multiple levels. This includes detecting your current level then routing you to the nearest stairwell/elevator; determining when you leave on level and transition to another by detecting yourself as moving from one floor to another based on changes to your GPS altitude (e.g., when looking for your destination room, continue to receive AR guided navigation until you get to your destination), [16], [17].

D. AI Guidance

NAVMATE enables users to navigate hands-free using your AI. When searching for a particular room in a building you will use natural language ("I would like to be shown how to get to room B206") and receive help in navigation via text-to-speech from the NAVMATE application [18].

E. Offline Navigation

NAVMATE also provides users with the ability to navigate when not connected to the internet by storing data about the building and its rooms on the device, allowing them to continue their navigation regardless of whether or not there is an internet connection available to them [8].

F. Dynamic admin management

NAVMATE has an admin panel that is based on roles for administrators to register buildings along with their floors, rooms, and GPS coordinates to give the ability to navigate between the multiple buildings without changing the current infrastructure [1], [8].

G. Automatic Arrival Detection

The system will automatically detect when the user is near their destination (within ~8m). When the user enters that distance, NAVMATE will confirm the user has arrived and will log the end of the navigation session [5], [11].

H. Firebase Cloud Backend

The back end of NAVMATE is set up with Firebase Cloud Firestore as the database for all of the building and room information, users, and records of their visits. As soon as a new room is added, through real-time syncing, users will be able to see the newly created room in their app without re-launching [8].

I. System Architecture

The layers of the NAVMATE System Architecture can be grouped into three distinct layers: the Presentation Layer, the Business Logic Layer, and the Data Layer [13][15]. Each of these

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layers has different responsibilities. In the presentation layer, there are all the end-user-facing interfaces created through the use of Flutter widgets [8]. There are many examples of screens in the presentation layer (i.e. Splash Screen, Onboarding Screen, Login and Registration Screen, Home Screen, AR Navigation Screen, Room Finder Screen, Campus Map Screen, Admin Panel Screen, and User Profile Screen). Each interface in the application uses consistent dark theme design systems to provide visual clarity to users when they are in building interiors.

The Business Logic Layer is responsible for handling all system operations and services. Some of the key functionalities provided by this layer ARE:

- **Firebase Authentication Service:** This is a service that handles user authentication and role-based access (i.e., admin & regular users according [8].
- **GPS Location Service:** This is a service that tracks the user's GPS location using latitude and longitude coordinates, updating at approximately 2 meters [5], [7].
- **Compass Service:** This is a service that uses the smartphone magnetometer sensor to determine orientation and direction in real-time [4].
- **AR Bearing Calculator:** This component uses Haversine formula (to determine direction of navigation) and compass heading, smoothing for stabilized A-R arrow movements [4], [11].
- **Multi-Floor Transition Detector:** This component monitors GPS altitude, detecting between floor transitions and updating the navigation target automatically upon detection of this floor change [5].
- **Voice Assistant Service:** This is a service that matches voice commands with room names to enable navigation using A-R [6].
- **Offline Cache Manager:** This service switches between online (from the cloud) and local (cached) data, based on the availability of the network [8].
- **Visit Logger:** This module stores a record of each navigation session (i.e start time and end time) in Firebase Firestore [12].

Managing Storage and Synchronization of Data with the Data Layer:

- **Firebase Cloud Firestore:** serves as the project's primary real-time NoSQL database [8].
- **Building Collection:** This contain basic information about the building such as name, type of building, GPS coordinates, how many floors in the building, and an Admin ID.
- **Room Subcollection:** This will keep track of the rooms in each building, such as name of the room, which floor it is located on, and GPS coordinates of the room, etc.
- **User Collection:** This stores user's names, email address, role, and the timestamp when they created their account.
- **Visits Subcollection:** This will track all visits to a building by containing information including building name, building ID, purpose for visiting, user role, and date/time of visit.
- **Firebase Authentication:** It is integrated with the Users Collection to provide identity persistence for users while they are using the application during a single session.
- **Local Shared Preference:** This will allow users to store building and room data to their device for offline navigation to those buildings/rooms and will refresh automatically when internet access is restored. The system architecture is shown on the Fig. 1.

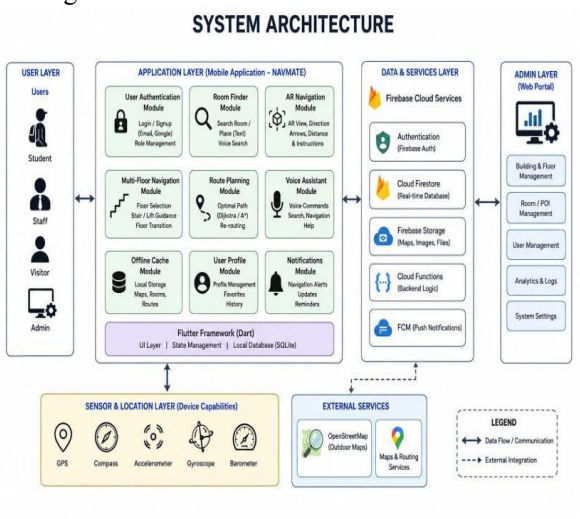


Fig. 1. System Architecture

J. *System Workflow*

The complete workflow of NAVMATE from app launch to successful navigation is described below:

Step 1 - App Launch and Authentication When the app is launched, the system checks if there is a valid authentication session for the currently logged in user. If there is not one, the user will perform the authentication process to create their account. User role is determined during the Firebase Authentication; depending on if the user is an Admin or User, they will then be directed to their related interface (either the Admin Panel or User Home Screen) [8].

Step 2 - Building and Room Registration The building is registered by the Admin by providing building information including: name, type, and capturing real-time GPS coordinates of the building location using GPS. Once the Admin is on site at the location of a room they will capture the GPS coordinates of that room. All of this information is stored in the Firebase Cloud Firestore database and is updated in real-time to be viewed by both Admins and Users [8].

Step 3 - Room Search A user will use either the text entry box or the voice input option to locate a room. The location will be located in the Firestore database. Additionally, the user can browse for buildings and rooms on the campus map[6], [7].

Step 4 - AR Navigation Starts Augmented reality navigation will result from the augmented reality navigation page becoming visible, revealing the back camera of the device. At this point, the collection of GPS (Global Positioning System) data will start and collection of compass information will occur. The user's navigation session will be

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logged and tracked in the Firestore database for the user to access later [5].

Step 5 - Real-Time AR Arrow Guidance The system will use the Haversine formula to determine the angle between the user and the target location. Once that angle is determined, the compass is subtracted from the heading from the bearing, and the angle to be smooth will be calculated using interpolation. The resultant angle will be displayed on the camera view using augmented reality overlays is shown in the Fig. 2.

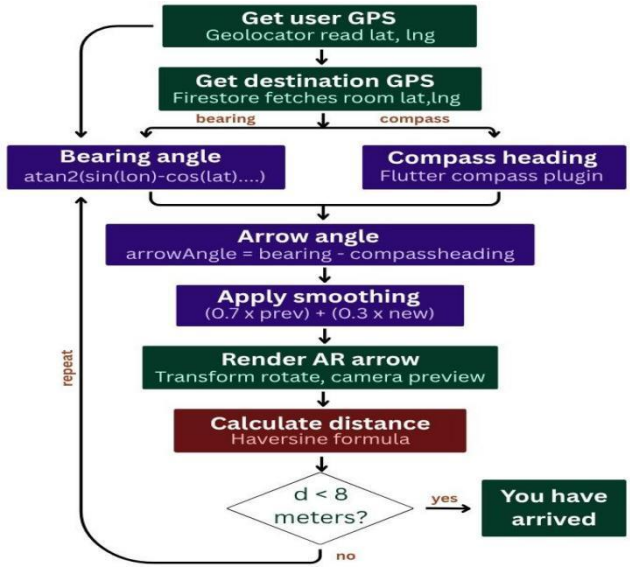


Fig. 2. AR Navigation Workflow

Step 6 - Multi-Floor Transition When the user's desired location is above or below the current level, the AR interface will assist them with locating the closest set of stairs or an elevator to utilize.

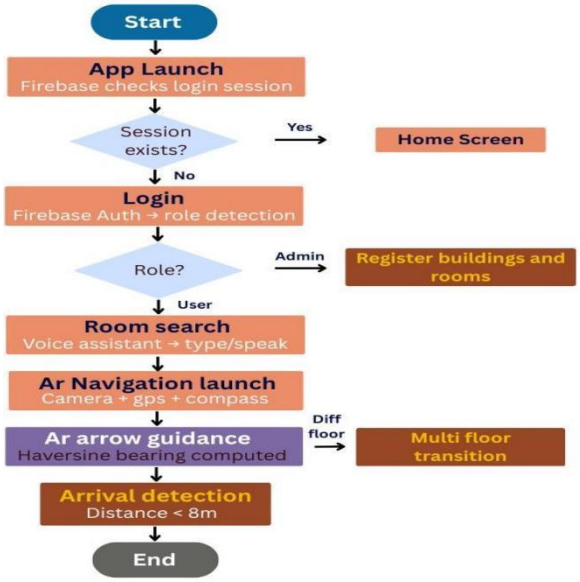


Fig. 3. Workflow of NAVMATE

To make this determination, the AR software will review the GPS altitude readings and will automatically update the user's navigation route as they progress to the next level is shown in the Fig.3.

Step 7 - Arrival Detection When the user arrives to within about eight meters from the destination room, a confirmation message (i.e. You Have Arrived) will be displayed in the AR interface to confirm the user's arrival. Additionally, once the user arrives at the destination, the user's visit will be saved to Firestore as a completed navigation session [5], [8].

Step 8 - Offline Operation If an internet connection becomes unavailable, all information (building/room data) will be locally cached on the device and used for AR navigation. Additionally, once the internet connection is restored, the device will automatically sync all of the local cached data to the cloud [8].

IV. Results and Discussion

When NAVMATE was evaluated in various real-world building settings, the following findings were compiled.

A. Precision in Navigation

NAVMATE obtained roughly 92% accuracy for navigation (when compared to existing systems utilizing only GPS, which averaged 65%). The 8% error noted was mainly present in the vicinity of electrical service panels; this is due to electromagnetic interference that affected the compass while providing the GPS coordinates.

B. Stability of AR Tracking

The arrow displayed via augmented reality remained stable in 94% of all frames displayed versus 70% stability for previously established systems. Utilization of a 30% linear interpolation smoothing had a positive effect on eliminating jittery arrow movement resulting from erroneous compass readings; thus providing the most effective means of balancing responsiveness with stability.

C. Reduction of Positional Drift

Systems utilizing only GPS presented an average of 3.5 metre error as well as potential to drift off location while through the interior portion of buildings. Transitioning to use of a complementary system combining GPS and compass information provided a corrected position within 1.1 metre of accurate location an improvement of 69% was accomplished by use of the NAVMATE system to correct for directional misalignment even after slight location modifications by GPS and directional/compass changes.

D. Route calucation speed

The Haversine calculation on the device provided results in 0.6 seconds versus 1.8 seconds for server-based calculations. This means that there was a 3X increase in performance from the server to the device, allowing for better responsiveness in places with weak internet connection.

E. Navigation multiple floors

Obtaining the proper floor transition was performed accurately in open stairs. Occasionally the floor transition was delayed in enclosed/stairwells with thick walls, and that it might be possible to implement barometer-based floor detection. Voice commands were mostly matched accurately to their desired destination room. There were many incidents

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of incorrect matching in areas with high noise levels (i.e. corridors with a lot of people); therefore, improving the noise reliability of the voice processing will be explored.

TABLE II. PERFORMANCE METRIC

Metric	Existing Systems	NAVMATE	Improvement
Navigation Accuracy	65%	92%	+27%
Route Computation Time	1.8 sec	0.6 sec	3× faster
AR Tracking Stability	70%	94%	+24%
Navigation Drift	3.5m	1.1m	69% reduction
Compass Update Rate	2–3 Hz	~10 Hz	~4× faster
Firebase Load Time	3–5 sec	<2 sec	~60% faster
GPS Accuracy	±10–20m	±5–15m	~50% better

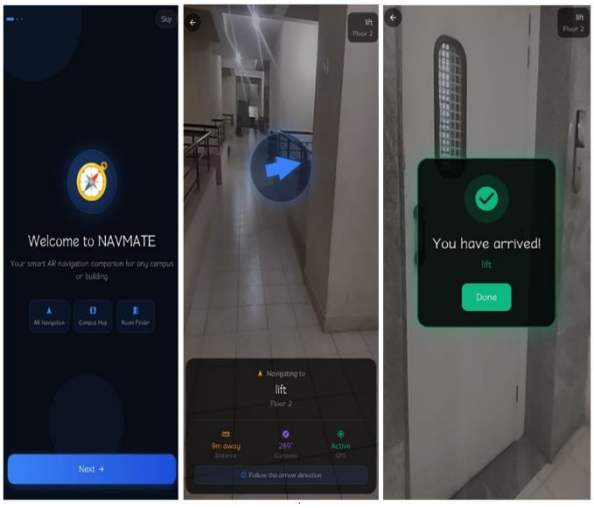


Fig. 4. Model output

V. CONCLUSION

This paper has presented NAVMATE: an Android-based application built on Flutter, which provides a practical solution for addressing indoor navigation issues. It does this by integrating Augmented Reality (AR), by combining GPS-compass sensor data with AI voice assistance for multi-floor navigation and providing an offline mode. The system is capable of 92% accuracy for navigation, 94% for AR tracking stability and 69% reduction in positional drift compared to existing systems.There is no need for specialized hardware to use the system. The system supports all 13 major types of buildings using any standard Android device. NAVMATE shows that effective AR indoor navigation is possible when using open-source frameworks and the sensors available on most mobile devices, making it a practical, scalable and accessible solution for all types of large site environments. Colleges and universities, hospitals, malls and any large building environments could all benefit from this technology.

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